

Money, Banks, and the Bank of Canada

ECON 101 – Macroeconomics

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Chapter Outline

- **10.1** What Is Money, and Why Do We Need It?
- 10.2 How Is Money Measured in Canada Today?
- 10.3 How Do Banks Create Money?
- 10.5 The Quantity Theory of Money

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10.1 Learning Objective

Goal

Define money and discuss the four functions of money.

- **Money:** any asset people generally accept in exchange for goods/services or to settle debts.
- Money is one of humanity's most important inventions:
 - enables specialization and trade,
 - supports development of complex economies.
- **Canada today:** physical cash accounts for <10% of transactions; the rest are digital (debit, Interac, credit, e-transfer).

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Life Without Money: The Problem with Barter

- **Barter:** trading goods/services directly for other goods/services.
- Requires a **double coincidence of wants**:
 - You must have what I want, AND I must have what you want, at the same time and place.
- **Why barter breaks down:**
 - A Prince George lumber worker wants a coffee. She offers boards.
 - Tim Hortons doesn't want boards.
 - *Barter fails.* Money solves this.
- In a complex economy with millions of goods, barter requires **millions** of exchange ratios. Money reduces this to just **one** price per good.

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From Barter to Commodity Money to Fiat Money

1. Barter

Direct exchange of goods.
Problem: double coincidence of wants.

*"I have wheat, you have fish.
Do you want wheat?"*

2. Commodity Money

Goods used as money (gold, silver, beaver pelts in early Canada).

Has intrinsic value.

Problem: heavy, indivisible, supply varies.

3. Fiat Money

Paper currency authorized by the Bank of Canada.

No intrinsic value.

Value based on trust and legal tender status.

Flexible supply.

- **Canadian history:** beaver pelts were used as money by early European settlers trading with Indigenous peoples. The *point blanket* trade served as a primitive unit of account.

The Four Functions of Money

1. Medium of exchange

- Accepted by a wide variety of parties for goods/services.
- Eliminates the need for double coincidence of wants.
- **Example:** paying for coffee with a \$5 bill or Interac.

2. Unit of account

- Standard way of measuring the value of goods.
- All Canadian prices quoted in **Canadian dollars (CAD)**.
- Without this, 1,000 goods $\Rightarrow$ 499,500 bilateral exchange ratios!

3. Store of value

- Transfers purchasing power from today to the future.
- **Inflation** reduces this function (why the BoC targets 2% inflation).

4. Standard of deferred payment

- Used to settle debts paid back in the future (mortgages, car loans, bonds).

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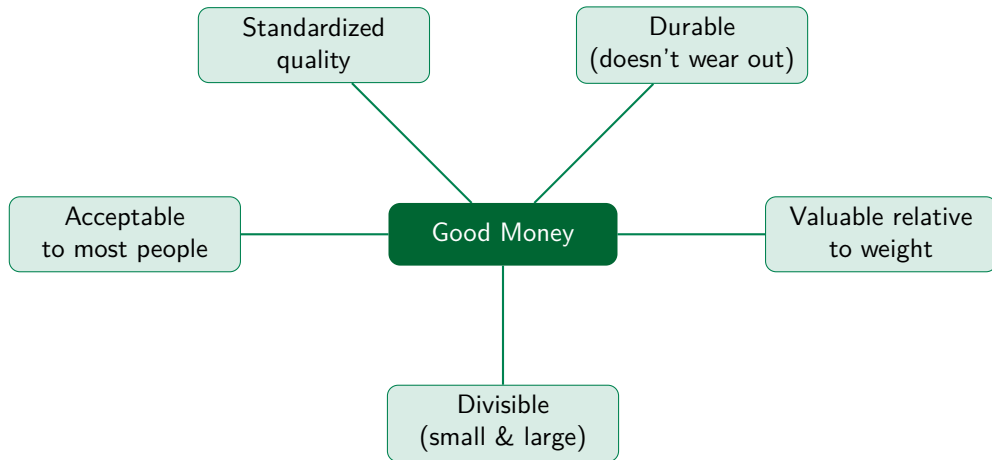
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What Makes a Good Money?



10.2 Learning Objective

Goal

Discuss the definitions of the money supply used in Canada today.

- Different assets vary in **liquidity** (ease of conversion to spending).
- The Bank of Canada defines multiple **monetary aggregates**, from narrow to broad.
- **2024 approximate values (Bank of Canada):**
 - M1+: $\approx$ \$1.2 trillion
 - M2: $\approx$ \$2.3 trillion
 - M2+: $\approx$ \$3.5 trillion

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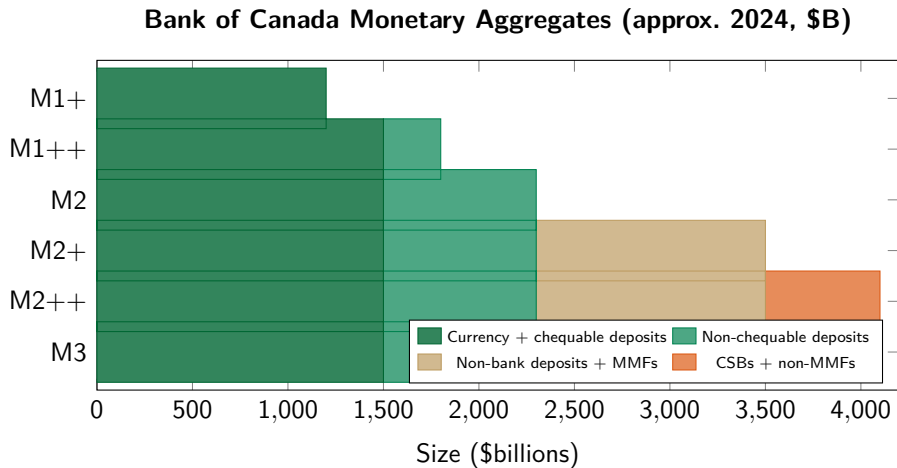
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Figure 10.1 – Canada's Monetary Aggregates (2024)



Source: Bank of Canada Banking and Financial Statistics, 2024. Values approximate.

Canada's Money Supply Measures Explained

Measure	What It Includes
M1+	Currency in circulation + chequable deposits at banks, trust companies, credit unions. <i>Most liquid.</i>
M1++	M1+ + non-chequable notice deposits.
M2	Currency + personal and non-personal deposits at banks.
M2+	M2 + deposits at trust/mortgage companies, credit unions (<i>caisses populaires</i>), MMFs, life insurance annuities.
M2++	M2+ + Canada Savings Bonds + non-money market mutual funds. <i>Broadest.</i>
M3	M2 + non-personal term deposits + foreign-currency deposits.

- For most macroeconomic analysis, **M1+** is most relevant (transactions money).

10.3 Learning Objective

Goal

Explain how banks create money.

- **How can banks create money?**
 - They accept deposits and make loans.
 - Loans create new deposits — the money supply **expands**.
- **Canada's major banks (Big Six):**
 - Royal Bank (RBC), TD Bank, Scotiabank, BMO, CIBC, National Bank.
 - Combined assets $\approx$ \$8 trillion (2024) — one of the most concentrated banking systems in the world.
- Banks in Canada are not legally required to hold a specific reserve ratio (unlike the U.S.).
- They set their own **desired reserve ratio** (r_d) based on liquidity needs.

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Bank Balance Sheets: A T-Account Example

- You deposit \$1,000 in cash at RBC.

Royal Bank — T-Account

Assets

Reserves +\$1,000

Liabilities

Deposits (chequing) +\$1,000

- Note:** currency in circulation fell by \$1,000, but chequing deposits rose by \$1,000.
- Total money supply is **unchanged so far**.
- But now RBC can lend!

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From Deposits to Loans: Creating New Money

- Suppose RBC keeps 10% ($r_d = 0.10$) as reserves and lends **\$900**.

RBC — After Making Loan

Assets

Reserves +\$100
Loan +\$900

Liabilities

Deposits +\$1,000

- The \$900 loan is spent by the borrower and deposited at TD Bank.
- TD now has \$900 in new deposits, keeps \$90, lends \$810.
- \$810 ends up at Scotiabank — and the process continues.
- This is the **money creation process**.

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From Deposits to Loans: Creating New Money

- Suppose RBC keeps 10% ($r_d = 0.10$) as reserves and lends \$900.

RBC — After Making Loan

Assets

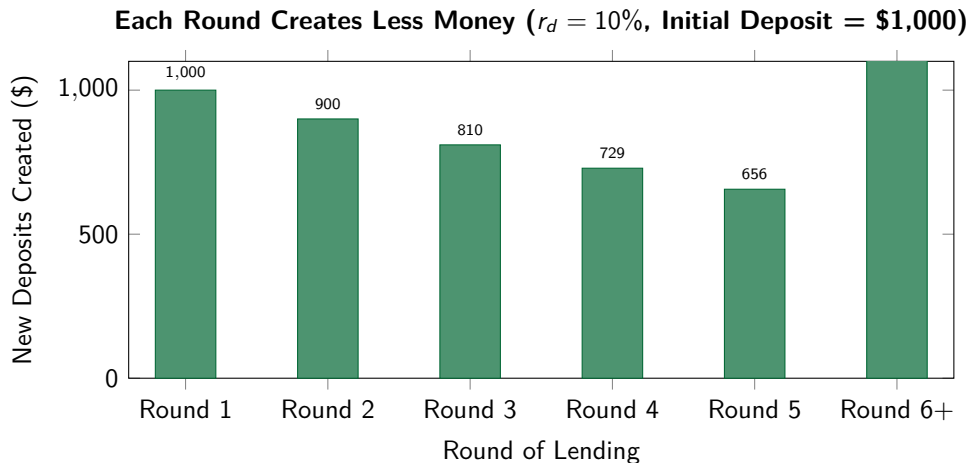
Reserves +\$100
Loan +\$900

Liabilities

Deposits +\$1,000

- The \$900 loan is spent by the borrower and deposited at TD Bank.
- TD now has \$900 in new deposits, keeps \$90, lends \$810.
- \$810 ends up at Scotiabank — and the process continues.
- This is the **money creation process**.

Figure 10.2 – Multiple Rounds of Money Creation



Total deposits = $\frac{1}{r_d} \times \$1,000 = \frac{1}{0.10} \times \$1,000 = \$10,000$. Banking system created \$9,000 of new money.

The Simple Deposit Multiplier

- **Simple deposit multiplier:**

$$\text{Total deposits} = \frac{1}{r_d} \times \Delta \text{Reserves.}$$

$$\text{Simple deposit multiplier} = \frac{1}{r_d}.$$

- **Examples:**

- $r_d = 10\%$: multiplier = 10.
- $r_d = 5\%$: multiplier = 20.
- $r_d = 20\%$: multiplier = 5.

- **Real-world multiplier is smaller** because:

1. Banks hold excess reserves (especially during crises).
2. Some loans are held as cash, not redeposited.

- **Canadian example (2020):** BoC emergency lending boosted bank reserves, but banks were cautious — real multiplier stayed low as lending was slow to respond.

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10.5 Learning Objective

Goal

Explain the quantity theory of money and use it to explain how high inflation occurs.

- The **quantity equation** (Irving Fisher, early 1900s):

$$M \times V = P \times Y,$$

where:

- M = money supply,
 - V = velocity of money (how often each dollar is spent per year),
 - P = price level,
 - Y = real GDP.
- Note: $P \times Y$ = **nominal GDP**.

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The Quantity Theory: Key Assumptions

- **Quantity theory of money:** assumes V is approximately constant in the long run.
- With V constant, changes in M must affect $P \times Y$.
- In growth-rate form:

$$g_M + g_V = \pi + g_Y.$$

- If $g_V \approx 0$:

$$g_M = \pi + g_Y \quad \Rightarrow \quad \boxed{\pi = g_M - g_Y}.$$

- **Implication:** inflation is a **monetary phenomenon** in the long run.
 - If money grows faster than real output $\Rightarrow$ inflation.
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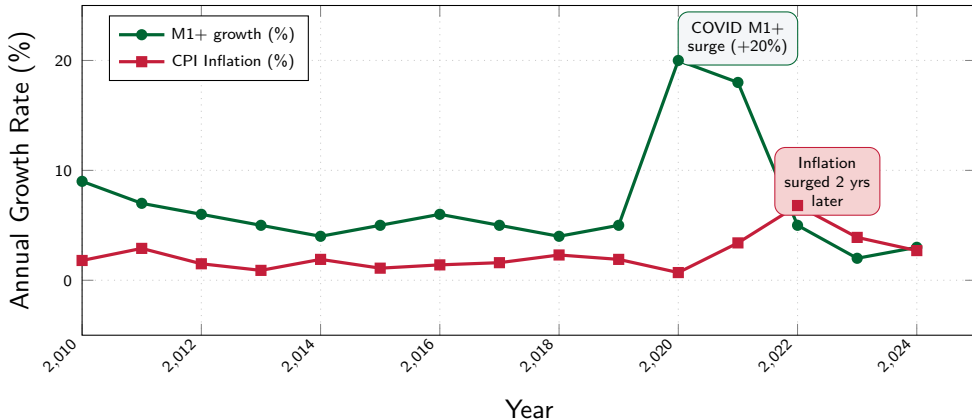
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Figure 10.3 – Money Growth and Inflation (Canada 2010–2024)

Money Supply Growth Led Inflation with a Lag (Canada)



Sources: Bank of Canada; Statistics Canada. M1+ and CPI data, annual averages.

Hyperinflation: Money Growth Gone Wrong

- **Hyperinflation:** inflation exceeding 50% per month.
- Caused by governments printing money to finance large budget deficits.
- **Historical examples:**
 - **Germany 1923:** prices doubled every 3.7 days — workers demanded to be paid twice daily and immediately spent wages.
 - **Zimbabwe 2008:** inflation reached $5 \times 10^{10}\%$ per month; the government issued 100 trillion dollar bills.
 - **Venezuela 2018:** $\approx 1,000,000\%$ annual inflation; many Venezuelans switched to USD or Bitcoin.
- **Lesson for Canada:** Bank of Canada independence and the 2% inflation target prevent excessive money printing. This institutional framework matters enormously.

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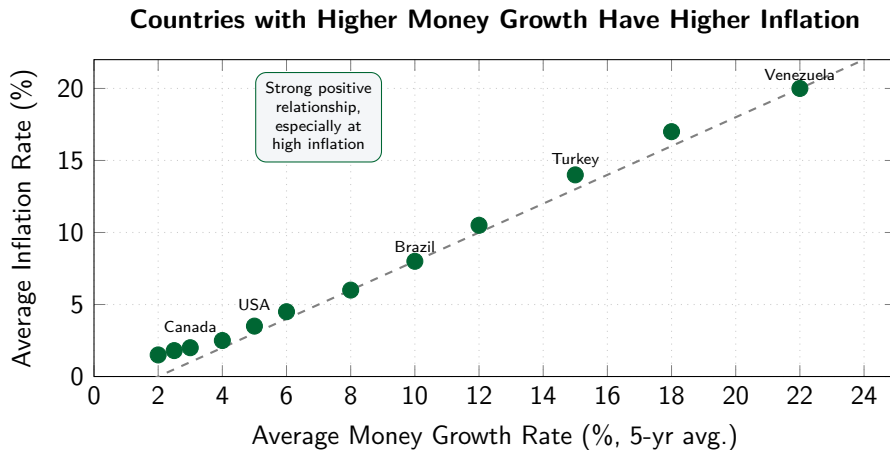
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Figure 10.4 – Quantity Theory: Cross-Country Evidence



Schematic illustration based on cross-country monetary data. High-inflation countries have high money growth.

Quantity Theory: Main Takeaways

- The quantity theory is primarily a **long-run** theory.
- Sustained inflation requires sustained money growth exceeding real GDP growth.
- Canada's COVID experience:
 - M1+ grew $\approx 20\%$ in 2020–21 (emergency pandemic liquidity).
 - Real GDP growth was modest ($\approx 5\%$ recovery).
 - Result: excess money growth contributed to the 2021–22 inflation surge.
 - Source: Bank of Canada, Banking and Financial Statistics.
- Bank of Canada's response: reduce money supply growth AND raise overnight rate — consistent with both the quantity theory and the AD-AS framework.

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Summary

- **Money** is any widely accepted asset. It performs four functions:
 - medium of exchange, unit of account, store of value, standard of deferred payment.
- Canada uses fiat money; the Bank of Canada controls the money supply.
- Monetary aggregates (M1+, M2, M2++) measure different degrees of liquidity.
- Banks **create money** by accepting deposits and making loans:

$$\text{Simple deposit multiplier} = \frac{1}{r_d}.$$

- Real-world multiplier is smaller due to excess reserves and cash holdings.
- The **quantity theory** ($\pi = g_M - g_Y$) predicts that persistent inflation requires persistent excess money growth.
- Canada's 2020–22 M1+ surge contributed to the worst inflation since the 1980s — tamed by the BoC's 2022–24 rate hiking cycle.

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- Canada's 2020–22 M1+ surge contributed to the worst inflation since the 1980s — tamed by the BoC's 2022–24 rate hiking cycle.

Thank you!

Questions?